Step by step from your equations: where the extra $\log\frac{\vee}{k^+}$ terms of Group III Rows III, V, VI and Group II Row IV come from; and the μ bookkeeping (running coupling, DGLAP)

Your notation throughout. Your plus prescription:

$$\int dp^+ \left[\frac{1}{p^+}\right]_+ f(p^+) \equiv \int dp^+ \frac{f(p^+)-f(0)}{p^+}, \qquad \int \frac{d\xi}{\xi^2}\,\frac{\xi}{(1-\xi)_+}\,g(\xi) \equiv \int \frac{d\xi}{\xi(1-\xi)}\,\big[g(\xi)-g(1)\big]. \tag{0.1}$$

Your `BFKL.tex` calls the Group III rows "III, IV, V"; in `Group_III.tex` they are Rows III, V, VI (Row IV has only four ρ's). Equation numbers "your (1.x)" are those of your files as they compile.

Your "first part of the calculation". The `.tex` file you mention did not reach me: the last files I received are `Group_I/II/III.tex` and `4rho.tex` (2.30.24 AM). Please upload it again and I will check it against the three Groups.

Part A. The extra terms in line 1

1 The answer in one paragraph

Nothing is added to your cross section. Every "extra" term is a piece of your line 1, the p^+ (resp. ξ) integral that runs up to $\vee - k^+$ (resp. down to $\frac{k^+}{\vee}$). It is written out because it grows like $\log\frac{\vee}{k^+}$. There are three of them in each row:

(i) The $\log\frac{\vee}{k^+}$ in your line 2, from $\log(\frac{\vee}{k^+}-1) = \log\frac{\vee}{k^+} + \log(1-\frac{k^+}{\vee})$.

(ii) Minus the same, hidden in the subtraction $-f(0)$ of $\left[\frac{1}{p^+}\right]_+$ in line 1. (i) and (ii) cancel exactly. So the $\log\frac{\vee}{k^+}$ you see in line 2 is *not* a $\vee$ dependence of the row: before your plus prescription, Eq. (your 1.12), there is no such logarithm (Step 3).

(iii) The term (B), hidden in the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term of line 1. This term has no $\frac{1}{p^+}$, but after the y' integral it becomes *exactly* $\frac{k^+}{p^++k^+}\times$ (a p^+-independent integral) + (a part that falls faster). Then $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+} = \log\frac{\vee}{k^+}$ (Step 4). This is the genuine BFKL term of the row.

In Group II Row IV, (i) and (ii) are the same with ξ. (iii) is H_4, from the $\log(y'-y)^2$ of the pole bracket in the $\frac{1-\xi}{\xi}$ term (Sec. 5).

2 Group III Row V, step by step

Step 1: your equation after the w' integral.

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ \left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)} - \frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\ \Big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^e(x)\rho^d(y)\Big]\end{aligned} \tag{your (1.12)}$$

After the Wilson lines are simplified (your Eq. (1.14)), the colour factor is

$$\mathcal{C}(y') = \Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big], \tag{2.1}$$

with $f^{c'ca}$ and $C \equiv \frac{1}{(2\pi)^3}\frac{ig^4 f^{c'ca}}{4\pi^4}\frac{1}{k^+}$ in front. Only $U^{e'c'}(y')$ depends on y'.

Step 2: names. Contract the Kronecker deltas of the three terms of the bracket. For each p^+,

$$F_a(p^+) = \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\, \frac{(y'-z)\cdot(x'-y')\,(y-w)\cdot(x-z)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\,\mathcal{C}(y') \qquad (\delta_{k'm}\delta_{ij}),$$

$$F_b(p^+) = \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\, \frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\,\mathcal{C}(y') \qquad (\delta_{jm}\delta_{ik'}),$$

$$F_c(p^+) = \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\, \frac{(y'-z)\cdot(y-w)\,(x'-y')\cdot(x-z)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\,\mathcal{C}(y') \qquad (\delta_{im}\delta_{jk'}). \tag{2.2}$$

Your Eq. (1.12), with the Wilson lines simplified, is then

$$\frac{d^3N_{\rm V}}{d^3k} = C\int_\Lambda^{\vee-k^+} \frac{dp^+}{2\pi}\left[\frac{F_a(p^+)}{p^+ + k^+} - \frac{F_b(p^+)}{p^+} - \frac{F_c(p^+)}{k^+}\right]. \tag{2.3}$$

Step 3: your plus prescription, and what it does to $\vee$. In your notation, exactly,

$$\int_\Lambda^{\vee-k^+} dp^+\,\frac{F_b(p^+)}{p^+} = \int_\Lambda^{\vee-k^+} dp^+\left[\frac{1}{p^+}\right]_+ F_b(p^+) + F_b(0)\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}, \qquad \int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+} = \log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big). \tag{2.4}$$

With $\int_\Lambda \to \int_0$ in the plus-prescribed term ($O(\Lambda)$), Eq. (2.3) is your Eq. (1.14):

$$\begin{aligned}\text{line 1} &= C\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\left[\frac{F_a(p^+)}{p^+ + k^+} - \frac{F_b(p^+)-F_b(0)}{p^+} - \frac{F_c(p^+)}{k^+}\right],\\ \text{line 2} &= -\frac{C}{2\pi}F_b(0)\left[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big)\right].\end{aligned} \tag{2.5}$$

($\frac{C}{2\pi} = \frac{1}{(2\pi)^3}\frac{ig^4 f^{c'ca}}{8\pi^5}\frac{1}{k^+}$, and $F_b(0)$, without the p^+ phase, is the integral of your line 2.)

The point. The left side of Eq. (2.4) does *not* grow like $\log\vee$: its integrand $\frac{F_b(p^+)}{p^+}$ falls at least like $\frac{1}{p^{+2}}$ (Step 4). The right side has $+F_b(0)\log\frac{\vee}{k^+}$ explicitly. So the plus-prescribed integral must contain $-F_b(0)\log\frac{\vee}{k^+}$: it subtracts $\frac{F_b(0)}{p^+}$ all the way up to $p^+ = \vee - k^+$. Exactly: since $\frac{1}{p^+} = \frac{1}{p^+ + k^+} + \frac{k^+}{p^+(p^+ + k^+)}$ and $\int_0^{\vee-k^+}\frac{dp^+}{p^+ + k^+} = \log\frac{\vee}{k^+}$,

$$\int_0^{\vee-k^+} dp^+\,\frac{F_b(p^+)-F_b(0)}{p^+} = \int_0^{\vee-k^+} dp^+\left[\frac{k^+\big(F_b(p^+)-F_b(0)\big)}{p^+(p^+ + k^+)} + \frac{F_b(p^+)}{p^+ + k^+}\right] \;-\; F_b(0)\,\log\frac{\vee}{k^+}. \tag{2.6}$$

This is identity (I2). Both integrands on the right fall like $\frac{1}{p^{+2}}$, so they are finite for $\vee\to\infty$. Put Eq. (2.6) into line 1, Eq. (2.5):

$$\begin{aligned}\text{(i)}\ &= -\frac{C}{2\pi}F_b(0)\log\frac{\vee}{k^+} \qquad (\text{the } \log\tfrac{\vee}{k^+} \text{ of line 2}),\\ \text{(ii)}\ &= +\frac{C}{2\pi}F_b(0)\log\frac{\vee}{k^+} \qquad \Big(\text{from } -\Big[\tfrac{1}{p^+}\Big]_+ \text{ in line 1}\Big), \qquad \text{(i)} + \text{(ii)} = 0.\end{aligned} \tag{2.7}$$

Written out, (i) is the first and (ii) the second $\log\frac{\vee}{k^+}$ line of Group III Row V in `Checks_JIMWLK_BFKL.pdf`.

Step 4: the y' integral, exactly. The two phases combine to one:

$$e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)} = e^{-ik\cdot(z-w)}\, e^{-iq\cdot(y'-z)}, \qquad q = t\,k, \quad t = \frac{p^+ + k^+}{k^+}. \tag{2.8}$$

In each F_n the integrand is $\frac{(y'-z)^m}{(y'-z)^2}G_n^m(y')$, where $G_n(y')$ collects $\frac{(x'-y')^{k'}}{(x'-y')^2}$ and $\mathcal{C}(y')$. Write, identically,

$$G_n(y') = G_n(z) + \big[G_n(y') - G_n(z)\big]. \tag{2.9}$$

For the $G_n(z)$ part the y' integral is a pure 2-D Fourier transform. With $\int d^2r\, e^{-iq\cdot r}\frac{r^m}{r^2} = -2\pi i\frac{q^m}{q^2}$ (from $\frac{r^m}{r^2} = \frac{1}{2}\partial^m\log r^2$ and $\int d^2r\, e^{-iq\cdot r}\log r^2 = -\frac{4\pi}{q^2}$):

$$\int_{y'} e^{-iq\cdot(y'-z)}\frac{(y'-z)^m}{(y'-z)^2} = -2\pi i\,\frac{1}{t}\,\frac{k^m}{k^2} = \frac{k^+}{p^+ + k^+}\int_{w'} e^{-ik\cdot(w'-z)}\frac{(w'-z)^m}{(w'-z)^2}. \tag{2.10}$$

This is identity (I3). It is exact; no large-p^+ limit is taken. Hence, for $n = a, b, c$,

$$F_n(p^+) = \frac{k^+}{p^+ + k^+}\,F_n^z + F_n^\Delta(p^+), \tag{2.11}$$

with

$$\begin{aligned}F_c^z &= \int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2(y-w)^2(x'-z)^2(x-z)^2}\,\mathcal{C}(z),\\ F_c^\Delta(p^+) &= \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\frac{(y'-z)^m(y-w)^m}{(y'-z)^2(y-w)^2}\,\frac{(x-z)^{k'}}{(x-z)^2}\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\mathcal{C}(y') - \frac{(x'-z)^{k'}}{(x'-z)^2}\mathcal{C}(z)\right],\end{aligned} \tag{2.12}$$

where $\mathcal{C}(z)$ is $\mathcal{C}(y')$ with $U^{e'c'}(y') \to U^{e'c'}(z)$, and similarly for a, b. $F_n^\Delta(p^+)$ falls faster than $\frac{1}{p^+}$:

- $G_n(y')-G_n(z)$ vanishes at $y'=z$, so the product with $\frac{(y'-z)^m}{(y'-z)^2}$ is bounded there, and its Fourier transform at $q=tk$ falls like $\frac{1}{q^2}\propto\frac{k^{+2}}{(p^++k^+)^2}$;
- at $y'=x'$ the factor $\frac{(x'-y')^{k'}}{(x'-y')^2}$ gives a $\frac{1}{q}$ term, but with the phase $e^{-iq\cdot(x'-z)}$, which oscillates in p^+. Its p^+ integral converges.

So $\int^\vee dp^+\frac{F_n^\Delta}{k^+}$, $\int^\vee dp^+\frac{F_n(p^+)}{p^++k^+}$ and $\int^\vee dp^+\frac{F_n(p^+)}{p^+}$ are all finite for $\vee\to\infty$. The only p^+ integrand of the row that falls like $\frac{1}{p^+}$ is the $G(z)$ part of the $-\frac{1}{k^+}$ term:

$$-\frac{C}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{k^+}{p^++k^+}\,F_c^z=-\frac{C}{2\pi}\,F_c^z\,\log\frac{\vee}{k^+}\qquad\text{exactly.}\tag{2.13}$$

Step 5: (iii), the term (B), written out. Eq. (2.13) with Eq. (2.12):

$$\begin{aligned}&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w',z,y,x,x'}e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\&\Big[U^{e'c'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]\end{aligned}\tag{(B)$_\vee$}$$

This is the blue (iii) line of Group III Row V in `Checks_JIMWLK_BFKL.pdf`.

Step 6: the row. Steps 3–5 give, exactly,

$$\frac{d^3N_{\rm V}}{d^3k}=\underbrace{-\frac{C}{2\pi}F_b(0)\log\frac{k^+}{\Lambda}}_{\text{JIMWLK}}+\underbrace{\text{(i)}+\text{(ii)}}_{=0}+\underbrace{(\text{B})_{\rm V}}_{\text{BFKL}}+R_{\rm V},\tag{2.14}$$

where $R_{\rm V}$ is line 1 with $\Big[\frac{1}{p^+}\Big]_+$ replaced as in Eq. (2.6), and with $G(y')\to G(y')-G(z)$ in the $-\frac{1}{k^+}$ term, plus line 2 with $\log(1-\frac{k^+}{\vee})$:

$$\begin{aligned}R_{\rm V}=&\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\&\Big[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\Big[\frac{1}{p^+}\Big]_+\Big(\Big[\frac{k^+}{p^+\,(p^++k^+)}\Big]_++\frac{1}{(p^++k^+)}\Big)\,\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\Big]\Big[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\&\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]\\&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\&\Big[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\Big]\Big[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\&\qquad-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{e'c'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Big]\times\\&\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]\\&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(1-\frac{k^+}{\vee}\Big)\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\&\Big[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]\end{aligned}\tag{2.15}$$

Every p^+ integrand in $R_{\rm V}$ falls at least like $\frac{1}{p^{+2}}$ or oscillates, so $R_{\rm V}$ is finite for $\vee\to\infty$. ($\Big[\frac{k^+}{p^+(p^++k^+)}\Big]_+F\equiv\frac{k^+[F(p^+)-F(0)]}{p^+(p^++k^+)}$.) This is the uniqueness argument: had we put a different amount of $\log\frac{\vee}{k^+}$ into the BFKL list, R would grow like $\log\vee$.

Numerical check (`bfkl_rows/exact_check.py`; line 1 of your new file is identical to the one checked there). Two configurations, $\vee=20k^+$, the p^+ integral done numerically, pointwise in the transverse points:

- line 1 = (rewritten line 1) + ($G(y')-G(z)$ term) + (ii) + ($G(z)$ term), Rows III(1), III(2), V, VI: $\le 3.7\times10^{-15}$;
- the $G(z)$ term after the Fourier identity (2.10) and the p^+ integral = (B): $\le 3.4\times10^{-15}$;
- the Fourier transforms themselves: $q\int_0^\infty J_1(qr)\,e^{-ar}dr$ at $a=10^{-2},3\times10^{-3},10^{-3}$ (must tend to 1): $q=0.7$: 0.9857, 0.9957, 0.9986; $q=1.9$: 0.9947, 0.9984, 0.9995. $2\pi\int_0^\infty rJ_0(qr)\log r^2\,e^{-ar^2}dr/(-\frac{4\pi}{q^2})$ at $a=4,2,1\times10^{-3}$, extrapolated to $a=0$ (must be 1): $q=0.7$: 1.0350, 1.0169, 1.0083 $\to$ 0.99971; $q=1.9$: 1.0045, 1.0022, 1.0011 $\to$ 1.00000.

3 Group III Rows VI and III (first part)

Your Eqs. (1.15)–(1.17) (Row VI) and (1.5)–(1.7) (Row III part 1) have the same three terms in the bracket. The steps are those of Sec. 2 word for word. Only the colour factor $\mathcal{C}(y')$ changes; it is $\left[U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')\right] \times [\dots]$ (and $f^{c'd'a}$). The results:

$$
\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\\
&\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]
\end{aligned}
\tag{(B)$_{\rm VI}$}
$$

$$
\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',w',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\\
&\times\left[U^{ac}(w)\,\{\rho^{c}(y),\rho^{b}(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^{d}(y),\rho^{e}(x)\}\right]
\end{aligned}
\tag{(B)$_{\rm III(1)}$}
$$

(i) and (ii) cancel as in Eq. (2.7).

4 Group III Row III, second part

Your Eq. (1.10), line 1:

$$
\begin{aligned}
&\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\\
&-\left[\frac{1}{p^+}\right]_+\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{p^+}{k^{+2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\\
&\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]
\end{aligned}
\tag{your (1.10), line 1}
$$

The bracket has six terms. After the y' integral, Eq. (2.11), each $G(z)$ part gets $\frac{k^+}{p^++k^+}$; the common factor is $\frac{1}{p^+(x-z)^2+k^+(x-w)^2} \sim \frac{1}{p^+}$. The large-$p^+$ power of each term, exactly:

term	coefficient $\times\frac{1}{p^+A+k^+B}\times\frac{k^+}{p^++k^+}$	$\int^\vee dp^+$
$\frac{\delta_{k'm}d_\perp p^+}{2(z-w)^2(p^++k^+)^2}[\dots]$	$\sim p^{+-3}$	finite
$\frac{p^+\delta_{k'm}}{k^+(p^++k^+)}[\dots]$	$\sim p^{+-2}$	finite
$\frac{\delta_{k'm}}{p^++k^+}[\dots]$	$\sim p^{+-3}$	finite
$-\left[\frac{1}{p^+}\right]_+[\dots]$	$F(p^+)$ part $\sim p^{+-3}$; $-F(0)$ part: (ii), as in Eq. (2.6)	(ii) $=-$(i)
$-\frac{p^+}{k^{+2}}[\dots]$	$\sim p^{+-1}$	$\log\frac{\vee}{k^+}$: (iii)
$-\frac{1}{k^+}[\dots]$	$\sim p^{+-2}$	finite

with $A=(x-z)^2$, $B=(x-w)^2$. For the $-\frac{p^+}{k^{+2}}$ term, the partial fractions (I5) are exact:

$$
-\frac{p^+}{k^{+2}}\,\frac{k^+}{p^++k^+}\,\frac{1}{p^+A+k^+B} = -\frac{1}{k^+A}\,\frac{1}{p^++k^+} + \frac{B}{A\,(p^++k^+)(p^+A+k^+B)}. \tag{4.1}
$$

The first term gives $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\frac{\vee}{k^+}$, i.e. (B); the second falls like $\frac{1}{p^{+2}}$ and stays in the remaining terms. Written out:

$$
\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\\
&\times\left[\frac{(w'-z)\cdot(z-w)}{(w'-z)^2\,(z-w)^2}-\frac{(w'-z)\cdot(x-w)}{2\,(w'-z)^2\,(x-w)^2}\right]
\end{aligned}
$$

$$\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\Big] \qquad ((\mathrm{B})_{\mathrm{III}(2)})$$

In the remaining terms (`Remaining_RC_DGLAP_finite.pdf`, Group III Row III part 2) the $-\frac{p^+}{k^+2}$ term becomes its $G(y')-G(z)$ part, and the second term of Eq. (4.1) is added:

$$+\ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}$$
$$\left[\frac{(x-w)^2}{(x-z)^2\,(p^++k^+)\,[p^+\,(x-z)^2+k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times$$
$$\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\Big] \qquad (4.2)$$

5 Group II Row IV

Step 1: your equations. With $\xi=\frac{k^+}{p^++k^+}$, your Eq. (1.19), and after the plus prescription, your Eq. (1.21):

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]$$
$$\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\ \frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\ \frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times$$
$$\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \qquad (\text{your } (1.21))$$

Call $g(\xi)$ the ξ-dependent part of the integrand of line 1 at fixed transverse points: $g(\xi)\propto e^{-ik\cdot(y'-y)\frac{1}{\xi}}$. Line 2 is $g(1)$ times $\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi(1-\xi)}=\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)$: your plus prescription, Eq. (0.1), in the ξ form of Eq. (2.4).

Step 2: (i) and (ii). $\frac{1}{\xi(1-\xi)}=\frac{1}{\xi}+\frac{1}{1-\xi}$ and $\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi}=\log\frac{\vee}{k^+}$ give, exactly,

$$\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,\frac{\xi}{(1-\xi)_+}g=\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\Big(\xi+\frac{\xi^2}{(1-\xi)_+}\Big)g\ -\ g(1)\log\frac{\vee}{k^+},\qquad\int\frac{d\xi}{\xi^2}\,\frac{\xi^2}{(1-\xi)_+}g\equiv\int\frac{d\xi}{1-\xi}\big[g(\xi)-g(1)\big]. \qquad (5.1)$$

The last term, times the prefactor of line 1, is (ii). It is your `BFKL.tex` line 1 for this row; it is exactly minus (i), the $\log\frac{\vee}{k^+}$ of line 2, poles included (5.5×10^{-16}).

Step 3: (iii), H_4. Which ξ integrand of line 1 falls like $\frac{1}{\xi}$ for $\xi\to\frac{k^+}{\vee}\to0$? The y' integral is a Fourier transform at $q=\frac{k}{\xi}$, large for small ξ. Write $H(y,y')=\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}[U^{e'c}(y')-U^{e'c}(x')][U^{ec}(y)-U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)=H(y,y)+[H(y,y')-H(y,y)]$.

- The constants of the pole bracket times $H(y,y)$: $\int_{y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}=(2\pi)^2\xi^2\delta^{(2)}(k)=0$ for $k\neq0$.
- $-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}$ times $H(y,y)$: $\int_{y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\log\frac{(y'-y)^2\mu^2}{4}=-\frac{4\pi\xi^2}{k^2}$ (μ drops out for $k\neq0$). With the measure $\frac{d\xi}{\xi^2}$, the three terms of the bracket give $\int d\xi\,\xi(1-\xi)$, $\int d\xi\,\xi$ (the ξ of Eq. (5.1)), $\int d\xi\,\frac{\xi}{1-\xi}\cdot$(subtracted): finite; and

$$\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{1-\xi}{\xi}=\log\frac{\vee}{k^+}-1+\frac{k^+}{\vee}. \qquad (5.2)$$

- $H(y,y')-H(y,y)$ vanishes at $y'=y$: its transform falls faster, finite.

So the only $\log\frac{\vee}{k^+}$ of line 1 besides (ii) is the $\frac{1-\xi}{\xi}$ term with $\log(y'-y)^2$ at $y'=y$. With $\frac{4\pi}{k^2}=\frac{1}{\pi}\int_{w,w'}e^{-ik\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$ and $y\to z$, its $\log\frac{\vee}{k^+}$ term is, exactly (identity (I4)),

$$\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\Big[U^{e'c}(z)-U^{e'c}(x')\Big]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \qquad (\log\tfrac{\vee}{k^+}H_4)$$

your `BFKL.tex` line 2 of this row. Its rest, $\big(-1+\frac{k^+}{\vee}\big)H_4\big|_{\log\to1}$, stays in the remaining terms. ($d_\perp=2$ here: no pole multiplies it.)

Step 4: the row. Group II Row IV $= \log\frac{k^+}{\Lambda}$ (your line 2, JIMWLK) + (i) + (ii) $+\log\frac{\vee}{k^+}H_4$ + a rest that is finite for $\vee\to\infty$; (i) + (ii) = 0.

Part B. The coefficient in front of $\log\mu^2$: running coupling and DGLAP

6 The five sources of μ

As your advisor says, μ enters through A3 and the shockwave through the loop, each with its c.c., and through the real term: five sources. Each comes in the combination $\hat{E}_r = \frac{2}{\epsilon} + \log\frac{r^2\mu^2}{c_0^2}$ ($c_0 = 2e^{-\gamma_E}$), so the coefficient of $\log\mu^2$ is the coefficient of $\frac{2}{\epsilon}$. In units of

$$\mathrm{LO} \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)$$

($\alpha_s = \frac{g^2}{4\pi}$; $L \equiv \log\frac{k^+}{\Lambda}$, $l_1 \equiv \log(1-\frac{k^+}{\vee})$, $X = x-y$, $\Delta = y'-y$):

A3, your Group I Eq. (1.2). Prefactor $\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+} = \frac{\alpha_sN_c}{4\pi}\times$(LO prefactor). With $w'\to y'$, $w\to y$ the colour factor is $-$(that of LO) and the bracket is $-\hat{E}_X$:

$$\mathrm{A3} + \mathrm{c.c.} = \frac{\alpha_sN_c}{4\pi}\,\mathrm{LO}\Big[(\hat{E}_X + \hat{E}_{X'})\Big(\frac{11}{3} - 2L + l_1\Big)\Big] + \ldots \quad\Rightarrow\quad \log\mu^2:\ \frac{\alpha_s}{\pi}\,\frac{N_c}{2}\Big(\frac{11}{3} - 2L + l_1\Big)\mathrm{LO}. \tag{6.1}$$

Shockwave through the loop, your Group I Eq. (1.32). Prefactor $\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+} = \frac{\alpha_sN_c}{2\pi}\times$(LO prefactor). With $y\to x$, $z\to w$ (dummy variables) the phase, kernel and colour factor are those of A3 with the opposite sign of the colour factor, and the bracket is $+\hat{E}$:

$$\mathrm{SW} + \mathrm{c.c.} = \frac{\alpha_sN_c}{2\pi}\,\mathrm{LO}\Big[(\hat{E}_X + \hat{E}_{X'})\Big(-\frac{11}{6} + 2L + 2\log(1-\tfrac{\Lambda}{k^+})\Big)\Big] \quad\Rightarrow\quad \log\mu^2:\ \frac{\alpha_s}{\pi}N_c\Big(-\frac{11}{6} + 2L\Big)\mathrm{LO}. \tag{6.2}$$

Real, your Group II Eq. (1.21). Prefactor $\frac{g^2}{4\pi^2}N_c\times$(LO prefactor)$= \frac{\alpha_s}{\pi}N_c$. The bracket $\big[-\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4} - 1\big] = -\hat{E}_\Delta + O(\epsilon)$ ($d_\perp = 2-\epsilon$). Line 1 with your $\frac{\xi}{(1-\xi)_+}$ is, by Eq. (5.1) and the $[0,1]$ plus distribution restricted to $[\frac{k^+}{\vee},1]$ ($\int_a^1\frac{d\xi\,h}{(1-\xi)_+} = \int_a^1\frac{h(\xi)-h(1)}{1-\xi}d\xi + h(1)\log(1-a)$, the $\overline{MS}$ one), $P^{(r)}_{gg,+}\otimes\mathrm{LO} - 2N_c\big(l_1 + \log\frac{\vee}{k^+}\big)\mathrm{LO}$:

$$\text{line 1}\Rightarrow\ \log\mu^2:\ -\frac{\alpha_s}{2\pi}\Big[\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi - 2N_c\Big(l_1 + \log\frac{\vee}{k^+}\Big)\mathrm{LO}\Big], \tag{6.3}$$

$$\text{line 2}\Rightarrow\ \log\mu^2:\ -\frac{\alpha_s}{2\pi}\,2N_c\Big(L + \log\frac{\vee}{k^+} + l_1\Big)\mathrm{LO}. \tag{6.4}$$

$P^{(r)}_{gg,+} = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big]$ is the real part of P_{gg}; it has no $\delta(1-\xi)$.

The coupling of LO. LO is $\propto g^2$. In $\overline{MS}$, $g^2 = g^2(\mu_R)\big[1 - \frac{\alpha_s\beta_0}{4\pi}\hat{E}\big]$, $\beta_0 = \frac{11}{3}N_c$:

$$\mathrm{CT} = -\frac{\alpha_s\beta_0}{4\pi}\,\hat{E}\,\mathrm{LO} \quad\Rightarrow\quad \log\mu^2:\ -\frac{\alpha_s}{\pi}\,\frac{\beta_0}{4}\,\mathrm{LO}. \tag{6.5}$$

Sum them up. Coefficient of $\log\mu^2$ in units of $\frac{\alpha_s}{\pi}$LO (`mu_check.py`; $P^* \equiv \int\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}\mathrm{LO}_\xi/\mathrm{LO}$):

source	β_0 $(\frac{11}{3})$	$\times L$	$\times\log\frac{\vee}{k^+}$	$\times l_1$	$\times P^*$
A3, Group I Row II part 1 + c.c., Eq. (6.1)	$11N_c/6$	$-N_c$	0	$N_c/2$	0
shockwave through the loop, Group I Row V 2b UV + c.c., Eq. (6.2)	$-11N_c/6$	$2N_c$	0	0	0
real, Group II Row IV line 1, Eq. (6.3)	0	0	N_c	N_c	$-1/2$
real, Group II Row IV line 2, Eq. (6.4)	0	$-N_c$	$-N_c$	$-N_c$	0
coupling of LO, Eq. (6.5)	$-11N_c/12$	0	0	0	0
sum of the five sources	0	0	0	$\frac{N_c}{2}$	$-\frac{1}{2}$
+ coupling of LO	$-\frac{11N_c}{12} = -\frac{\beta_0}{4}$	0	0	$\frac{N_c}{2}$	$-\frac{1}{2}$

So:

1. **β_0: the four sources of the single-gluon final state cancel exactly**: $\frac{N_c}{2}\cdot\frac{11}{3} - N_c\cdot\frac{11}{6} = 0$. Not only $\log\mu^2$: the pole, γ_E and $\log(x-w)^2$ cancel too, pointwise after the renaming (`dglap_rc/checks.json`: 0). This is what your advisor expected.

2. **$\times\log\frac{k^+}{\Lambda}$ (evolution): three sources**, A3 : shockwave : real $= -N_c : 2N_c : -N_c$, sum 0. They cancel only with the real term. Pointwise the poles are in the ratio $1:-2:1$ and cancel to 8.2×10^{-16} (`ll_poles.json`). These are the three sources of the evolution.

3. $\times \log \frac{\vee}{k^+}$: $+N_c$ (line 1, (ii)) and $-N_c$ (line 2, (i)) cancel: the BFKL terms have no μ.

4. What is left: $-\frac{\alpha_s}{2\pi}\log\mu^2\big[\int P^{(r)}_{gg,+}\otimes \mathrm{LO} + \frac{\beta_0}{2}\mathrm{LO}\big] = -\frac{\alpha_s}{2\pi}\log\mu^2\, P_{gg}\otimes \mathrm{LO}$ (DGLAP, with $P_{gg} = P^{(r)}_{gg,+} + \frac{\beta_0}{2}\delta(1-\xi)$), and $+\frac{\alpha_s}{2\pi}N_c\, l_1 \log\mu^2\,\mathrm{LO}$ (Q1, from A3 only; $l_1 \to 0$ for $\vee \to \infty$).

7 Why β_0 is not in the NLO terms, and whether Group II Row IV is running coupling

- **The NLO diagrams carry no** $\beta_0 \log\mu^2$, because of item 1: A3 $(+\frac{11}{3})$ and the shockwave through the loop $(2\times(-\frac{11}{6}))$ cancel, and the real term has no β_0. So at this order nothing in your rows fixes a scale for the coupling; in particular no $\beta_0 \log(x-w)^2$ survives.

- β_0 **is still in the result: through the coupling of LO.** LO is $O(g^2)$, so g^2 must be renormalized; Eq. (6.5) is the only β_0. Its pole is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of P_{gg}, which the real term lacks. After $\overline{MS}$ renormalization of g^2 at μ_R and of $D_{h/g}$ at μ_F, the β_0 term of the result is

$$\frac{\alpha_s\beta_0}{4\pi}\,\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO}\Big|_{\alpha_s(\mu_R)}, \tag{7.1}$$

  and the result is μ_R-independent: $\frac{d}{d\log\mu_R^2}\mathrm{LO}|_{\alpha_s(\mu_R)} = -\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}$. This is the running coupling: the coupling of LO runs, the NLO diagrams do not add to it.

- **Group II Row IV is not running coupling.** Its $\log\mu^2$ (Eqs. (6.3), (6.4)) multiplies $\int \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}(\xi)\mathrm{LO}_\xi$, a convolution in ξ with the *real* splitting function. It has no $\delta(1-\xi)$ part and no β_0: the plus distribution integrates the $\frac{\xi}{1-\xi}$ singularity against $g(\xi)-g(1)$, and its $\delta(1-\xi)$ coefficient is zero in the $\overline{MS}$ convention, Eq. (6.3). A running-coupling term would be $\beta_0\log\mu^2\times\mathrm{LO}$ (local in ξ). What Group II Row IV gives is the μ_F dependence of the fragmentation function (DGLAP), and $-2N_cL\log\mu^2$, which cancels against Group I (item 2).

- With both: $-\frac{\alpha_s}{2\pi}\log\mu^2\, P_{gg}\otimes\mathrm{LO}$ is exactly what $\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g} = \frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}$ cancels. The coupling counterterm completes $P^{(r)}_{gg,+}$ to the full P_{gg}.